Dataset Creation Plan

Project Initiation		
Project Title:	Premature Mortality in Jamaica: Assessing Trends, Patterns, Costs, and Socioeconomic Disparities	
Department:	Caribbean Institute for Health Research	
Unit:	Epidemiology Research Unit	
Project Objectives:		
	Thesis Objective 1: To describe trends and patterns in overall and cause-specific premature mortality and total mortality in Jamaica between 2010 and 2019.	
Principal Investigator (PI):	Trevor Ferguson	
Project Team Member(s) Responsible for Project Dataset Creation and/or Statistical Analysis and date joined (list all):		
	Tonya Campbell	2023-02-13
Other Project Team Members and date joined (list all):	*List all project team members*	
	Novie Younger-Coleman Marshall Tulloch-Reid Ian Hambleton Lizheng Shi	2023-02-13
DCP Creation Date and Author:	*Date DCP was finalized prior to Project Dataset(s) creation*	*Name of person who created the DCP*
	Date	***Name***
	May 24, 2024	Tonya Campbell

Dataset Creation Plan

Data	
Data Source	***Years (where applicable)***
Mortality	
Registrar General's Department Mortality Database	2010 – 2019
Jamaica Constabulary Force homicide counts, aggregated by age group and sex	2013 – 2019
Population	
STATIN's national mid-year population estimates (Demography Statistics Report 2019 + Population Statistics webpage https://statinja.gov.jm/Demo_SocialStats/PopulationStats.aspx)	2010 – 2019
WHO World Standard Population (2000-2025)	

Project Amendments and Reconciliation

DCP Amendment History (add additional rows as needed):

Date DCP amended	*Person who made the DCP amendment*	
Date	***Name***	***Amendment***
October 14, 2026	Tonya Campbell	Reconciliation of the SE method, Poisson models, and JCF analyses
July 20, 2026	Tonya Campbell	Adding formulas used to calculate standard error for crude/age-specific and age-adjusted rates
June 16, 2026	Tonya Campbell	Minor changes to align with Do File (removing references to 75-year threshold; revising info about cause of death groupings and statistical analysis)
Aug 4, 2025	Tonya Campbell	Adding a metric for median YPLL Amending cause of death groupings Adding statistical tests to compare the pct of premature death deaths and median YPLL by sex and cause
Nov 25, 2024	Tonya Campbell	Adding date that Stata Do File program was reviewed, and amending diagnosis codes included in cause of death groupings to align with groupings in the WHO MDB
Sept 19, 2024	Tonya Campbell	Updated data sources and indicator details
July 10, 2024	Tonya Campbell	Updated cause of death categories, age groupings, and study period, added appendix
May 27, 2024	Tonya Campbell	Updated cause of death categories, exclusions, study design, and analysis plan based on meeting with Prof Ferguson on May 27, 2024

Date Programs/DCP reconciled

The person(s) creating the dataset and/or analyzing the data are responsible for ensuring that the final DCP reflects the final program(s) when the project is completed

October 14, 2026

Study Details

Study Details	
Study Design	Descriptive ecological study
Study Period	January 1, 2010 to December 31, 2019
Time Interval	Annual

Denominators

Population Denominators	
Population	**Use STATIN's mid-year population estimates to determine:** • Population of Jamaica, 2010 to 2019 • Population of Jamaica by age group (0-14, 15-29, 30-39, 40-49, 50-59, 60-69 years), 2010 to 2019 ○ **Note:** this categorization will be used for reporting age-specific mortality • Population of Jamaica by age group (0-4, 5-9, 10-14, 15-19, 20-24, 25-29, 30-34, 35-39, 40-44, 45-49, 50-54, 55-59, 60-64, 65-69, 70-74, 75+ years), 2010 to 2019 ○ **Note:** this categorization will be used for age standardization • Population of Jamaica by sex, 2010 to 2019 • Population of Jamaica by age group (0-14, 15-29, 30-39, 40-49, 50-59, 60-69 years) and sex, 2010 to 2019 • Population of Jamaica by age group (0-4, 5-9, 10-14, 15-19, 20-24, 25-29, 30-34, 35-39, 40-44, 45-49, 50-54, 55-59, 60-64, 65-69, 70-74, 75+ years) and sex, 2010 to 2019

Indicators

Indicator Title	1. Incidence of Mortality
Data Sources	• **RGD Mortality Database**: To identify individuals who died during the study period • **STATIN Mid-Year Population Estimates:** To determine population
Denominator population description (Source, exclusions, time frame)	1) Population of Jamaica, overall (annually: 2010 to 2019) 2) Population of Jamaica, by age group (annually: 2010 to 2019) i. **Age group for 70-year referent:** 0-14, 15-29, 30-39, 40-49, 50-59, 60-69 years 3) Population of Jamaica, by sex (annually: 2010 to 2019) 4) Population of Jamaica, by age group and sex (annually: 2010 to 2019)
Numerator (Usually a subset of denominator; restricted as follows:)	Total number of people who died during the year of interest. Report: • Overall, by year • Age group, by year i. 0-14, 15-29, 30-39, 40-49, 50-59, 60-69 years • Sex, by year • Age group and sex, by year
Exclusions	1. Individuals with missing information regarding age at death 2. People who resided outside of Jamaica at the time of death (use the RESIDENCE_PARISH variable) **Note:** Report number excluded at each step
Other details (Multiplier for rates, stratifications, etc.)	1. Express all rates per 100,000 population. 2. Report results for all causes of death and stratified by cause (See Appendix for diagnosis codes): i. Maternal & Perinatal conditions ii. Infectious & Parasitic diseases iii. Cancer iv. Diabetes & Endocrine Disorders v. Circulatory Diseases vi. Chronic Respiratory Diseases vii. Oral & Digestive Diseases viii. Genitourinary Diseases ix. Neuropsychiatric Conditions x. External Causes xi. Other Causes
Age standardization	1. For rates reported overall and by sex, standardize to the age distribution of the WHO World Standard Population (2000-2025) a. To do this, in each year of the study period, multiply the age-specific mortality rate for each of the following age groups (0-4, 5-9, 10-14, 15-19, 20-24, 25-29, 30-34, 35-39, 40-44, 45-49, 50-54,

	55-59, 60-64, 65-69, 70-74, 75+ years) by the proportion of the WHO Standard Population in the same age group (i.e., the weight) b. Sum the products across all age groups to get the age-standardized mortality rate 2. Age standardization will be conducted for rates reported overall and by sex, for all causes combined and each major cause separately
Visualization	1. Figures of trends over time in all-cause mortality rates, crude: 1.1. Overall and by sex 1.2. By age group 2. Figures of trends over time in cause-specific mortality rates, crude: 2.1. Overall 2.2. Among females 2.3. Among males 3. Figures of trends over time in all-cause mortality rates, age-standardized: 3.1. Overall and by sex 4. Figures of trends over time in cause-specific mortality rates, age-standardized: 4.1. Overall 4.2. Among females 4.3. Among males 5. Table displaying all-cause and cause-specific mortality rates (crude and age-standardized) overall and by sex, 2010-2019 6. Table displaying all-cause and cause-specific mortality rates overall and by age group, 2010-2019

Indicator Title	2. Percentage of Deaths that are Premature
Data Sources	• **RGD Mortality Database**: To identify individuals who died prematurely during the study period • **STATIN Mid-Year Population Estimates:** To determine population
Denominator population description (Source, exclusions, time frame)	1) Total number of deaths, overall (annually: 2010 to 2019) 2) Total number of deaths, by sex (annually: 2010 to 2019)
Numerator (Usually a subset of denominator; restricted as follows:)	Total number of people who died *prematurely* (see threshold below) during the year of interest. As the primary analysis, use a threshold of *<70 years of age*. Report: • Overall, by year • Sex, by year
Exclusions	1. Individuals with missing information regarding age at death 2. People who resided outside of Jamaica at the time of death (use the RESIDENCE_PARISH variable) **Note:** Report number excluded at each step
Other details (Multiplier for rates, stratifications, etc.)	1. Report results for all causes of death and stratified by cause (See Appendix for diagnosis codes): i. Maternal & Perinatal conditions ii. Infectious & Parasitic diseases iii. Cancer iv. Diabetes & Endocrine Disorders v. Circulatory Diseases vi. Chronic Respiratory Diseases vii. Oral & Digestive Diseases viii. Genitourinary Diseases ix. Neuropsychiatric Conditions x. External Causes xi. Other Causes
Visualization	1. Figure of trends over time in the percentage of premature deaths, overall and by sex 2. Figure displaying the percentage of premature deaths by cause and sex in 2010 and 2019 3. Table displaying the percentage of premature deaths (overall and cause-specific mortality) overall and by sex, 2010-2019

Indicator Title	3. Total Years of Potential Life Lost (YPLL)
Data Sources	• **RGD Mortality Database**: To identify individuals who died prematurely during the study period • **STATIN Mid-Year Population Estimates:** To determine population
Denominator population description (Source, exclusions, time frame)	1) Population of Jamaica, overall (annually: 2010 to 2019) 2) Population of Jamaica, by age group (annually: 2010 to 2019) i. **Age group for 70-year referent:** 0-14, 15-29, 30-39, 40-49, 50-59, 60-69 years 3) Population of Jamaica, by sex (annually: 2010 to 2019) 4) Population of Jamaica, by age group and sex (annually: 2010 to 2019)
Numerator (Usually a subset of denominator; restricted as follows:)	Total number of years of potential life lost *(YPLL)* due to *premature* death (see threshold below) during the year of interest. As the primary analysis, use a threshold of *<70 years of age*. For each person, subtract the age at death from the threshold age. Then sum the years of potential life lost according to the following groupings, and report: • Overall, by year • Age group, by year i. 0-14, 15-29, 30-39, 40-49, 50-59, 60-69 years • Sex, by year • Age group and sex, by year
Exclusions	1. Individuals with missing information regarding age at death 2. People who resided outside of Jamaica at the time of death (use the RESIDENCE_PARISH variable) **Note:** Report number excluded at each step
Other details (Multiplier for rates, stratifications, etc.)	1. Express rates per 100,000 population 2. Report results for all causes of death and stratified by cause (See Appendix for diagnosis codes): i. Maternal & Perinatal conditions ii. Infectious & Parasitic diseases iii. Cancer iv. Diabetes & Endocrine Disorders v. Circulatory Diseases vi. Chronic Respiratory Diseases vii. Oral & Digestive Diseases viii. Genitourinary Diseases ix. Neuropsychiatric Conditions x. External Causes xi. Other Causes
Age standardization	1. For rates reported overall and by sex, standardize to the age distribution of the WHO World Standard Population (2000-2025). For the primary analysis (threshold of age 70

	years), recalculate the weights to restrict to the population <70 years of age. a. In each year of the study period, multiply the age-specific YPLL rate for each of the following age groups (0-4, 5-9, 10-14, 15-19, 20-24, 25-29, 30-34, 35-39, 40-44, 45-49, 50-54, 55-59, 60-64, 65-69 years) by the proportion of the WHO Standard Population in the same age group (i.e., the re-calculated weight) b. Sum the products across all age groups to get the age-standardized YPLL rate 2. Age standardization will be conducted for rates reported overall and by sex, for all causes combined and each major cause separately
Visualization	1. Figures of trends over time in all-cause YPLL rates, crude: 1.1. Overall and by sex 1.2. By age group 2. Figures of trends over time in cause-specific YPLL rates, crude: 2.1. Overall 2.2. Among females 2.3. Among males 3. Figures of trends over time in all-cause YPLL rates, age-standardized: 3.1. Overall and by sex 4. Figures of trends over time in cause-specific YPLL rates, age-standardized: 4.1. Overall 4.2. Among females 4.3. Among males 5. Figures of trends over time in YPLL rates by cause for each combination of age group and sex 6. Table displaying all-cause and cause-specific YPLL rates overall and by sex (crude and age-standardized), 2010-2019 7. Table displaying all-cause and cause-specific YPLL rates overall and by age group, 2010-2019

Indicator Title	4. Median Years of Potential Life Lost (YPLL) per Death
Data Sources	• **RGD Mortality Database**: To identify individuals who died prematurely during the study period
Numerator	The median number of years of potential life lost for each person at the age at death. As the primary analysis, calculate using a threshold of *<70 years of age*. For each person, subtract the age at death from the threshold age. Then calculate the median of the years of life lost per death and report: • Overall, by year • Sex, by year
Exclusions	1. Individuals with missing information regarding age at death 2. People who resided outside of Jamaica at the time of death (use the RESIDENCE_PARISH variable) **Note:** Report number excluded at each step
Other details (Multiplier for rates, stratifications, etc.)	1. Report results for all causes of death and stratified by cause (See Appendix for diagnosis codes): i. Maternal & Perinatal conditions ii. Infectious & Parasitic diseases iii. Cancer iv. Diabetes & Endocrine Disorders v. Circulatory Diseases vi. Chronic Respiratory Diseases vii. Oral & Digestive Diseases viii. Genitourinary Diseases ix. Neuropsychiatric Conditions x. External Causes xi. Other Causes
Visualization	1. Figure of trends over time in median YPLL, overall and by sex 2. Figure displaying median YPLL by cause and sex in 2010 and 2019 3. Table displaying all-cause and cause-specific median YPLL overall and by sex, 2010-2019

Indicator Title	5. Unconditional Probability of Dying Between Ages 30 And 70 Years from Non-Communicable Diseases
Data Sources	• **RGD Mortality Database**: To identify individuals who died prematurely during the study period • **STATIN Mid-Year Population Estimates:** To determine population
Measure	Unconditional probability of dying between the exact ages 30 and 70 years from circulatory diseases (I00-I99), cancer (C00-C97), diabetes (E10-E14), or chronic respiratory diseases (J30-J98) To calculate this measure, first generate age-specific death rates for the combined cause of death as: $${}_5M_x = \frac{Total\ deaths\ from\ four\ NCD\ causes\ between\ exact\ age\ x\ and\ exact\ age\ x+5}{Total\ population\ between\ exact\ age\ x\ and\ exact\ age\ x+5}$$ Next, translate the 5-year death rate to the probability of death in each 5-year age range as: $${}_5q_x = \frac{{}_5M_x * 5}{1 + {}_5M_x * 2.5}$$ Then calculate unconditional probability of death from age 30 to age 70 as: $${}_{40}q_{30} = 1 - \prod_{x=30}^{65}(1 - {}_5q_x)$$
Exclusions	1. Individuals with missing information regarding age at death 2. People who resided outside of Jamaica at the time of death (use the RESIDENCE_PARISH variable) **Note:** Report number excluded at each step
Visualization	1. Figure of trends over time, overall and by sex, all-cause and by cause 2. Table displaying all-cause and cause-specific probabilities, overall and by sex, 2010-2019

Analysis Plan

Analysis Plan
Tables
Table 1a. Trends in crude total and premature mortality in Jamaica, overall and by cause of death, 2010 to 2019 (70 year referent)
Table 2a. Trends in crude total and premature mortality in Jamaica, by sex and cause of death, 2010 to 2019 (70 year referent)
Table 3a. Trends in premature mortality in Jamaica, by age and cause of death, 2010 to 2019 (70 year referent)
Table 4a. Trends in premature mortality in Jamaica, by age, sex, and cause of death, 2010 to 2019 (70 year referent)
Table 5a. Trends in age-standardized total and premature mortality in Jamaica, overall and by cause of death, 2010 to 2019 (70 year referent)
Table 6a. Trends in age-standardized total and premature mortality in Jamaica, by sex and cause of death, 2010 to 2019 (70 year referent)
Analysis Plan

- For each indicator, calculate rates (expressed per 100,000 population) or percentages as appropriate. Perform calculations for the population overall, by age group, sex, age group and sex, as outlined above. Calculate the percentage change for all indicators.
- Calculate age standardized rates of total mortality and premature mortality (measured as total YPLL). Standardize according to the WHO World Standard 2000-2025
- Calculate the standard errors for crude and age-specific rates of YPLL (for use in the Joinpoint models) using the Poisson approximation as follows:
    1. **Crude and age-specific YPLL rates:**
        - For each relevant analytic stratum, square each individual YPLL value and then sum the squared values.
        - Divide the square root of this sum by the relevant population denominator: the population younger than 70 years for overall and sex-specific crude rates, or the corresponding age-group-specific population for age-specific rates.
        - Multiply the result by 100,000 to obtain the standard error per 100,000 population.
    2. **Directly age-standardized YPLL rates:**
        - Within each five-year age group, square each individual YPLL value and then sum the squared values.
        - For each age group, multiply the sum of the squared individual YPLL values by the square of the corresponding renormalized WHO World Standard Population weight.
        - Divide this value by the square of the corresponding age-specific population.
        - Sum the resulting variance components across all age groups, by year and cause of death as appropriate, to obtain the variance of the directly age-standardized YPLL rate.
        - Take the square root of this variance and multiply by 100,000 to obtain the standard error per 100,000 population.
- For the following indicators, use Joinpoint regression (with the standard errors calculated above) to estimate the average annual percent change in age-standardized rates of total YPLL and assess whether changes in trends are statistically significant:

All-Cause

1. Trends in premature mortality (YPLL rate) – overall
2. Trends in premature mortality (YPLL rate) – among females
3. Trends in premature mortality (YPLL rate) – among males
4. Trends in premature mortality (YPLL rate) – among 0-14 age group
5. Trends in premature mortality (YPLL rate) – among 15-29 age group
6. Trends in premature mortality (YPLL rate) – among 30-39 age group
7. Trends in premature mortality (YPLL rate) – among 40-49 age group

Analysis Plan

8. Trends in premature mortality (YPLL rate) – among 50-59 age group
9. Trends in premature mortality (YPLL rate) – among 60-69 age group

Maternal and perinatal conditions

10. Trends in premature mortality (YPLL rate) – overall
11. Trends in premature mortality (YPLL rate) – among females
12. Trends in premature mortality (YPLL rate) – among males
13. Trends in premature mortality (YPLL rate) – among 0-14 age group
14. Trends in premature mortality (YPLL rate) – among 15-29 age group
15. Trends in premature mortality (YPLL rate) – among 30-39 age group

Infectious & parasitic diseases

16. Trends in premature mortality (YPLL rate) – overall
17. Trends in premature mortality (YPLL rate) – among females
18. Trends in premature mortality (YPLL rate) – among males
19. Trends in premature mortality (YPLL rate) – among 0-14 age group
20. Trends in premature mortality (YPLL rate) – among 15-29 age group
21. Trends in premature mortality (YPLL rate) – among 30-39 age group
22. Trends in premature mortality (YPLL rate) – among 40-49 age group
23. Trends in premature mortality (YPLL rate) – among 50-59 age group
24. Trends in premature mortality (YPLL rate) – among 60-69 age group

Cancer

25. Trends in premature mortality (YPLL rate) – overall
26. Trends in premature mortality (YPLL rate) – among females
27. Trends in premature mortality (YPLL rate) – among males
28. Trends in premature mortality (YPLL rate) – among 0-14 age group
29. Trends in premature mortality (YPLL rate) – among 15-29 age group
30. Trends in premature mortality (YPLL rate) – among 30-39 age group
31. Trends in premature mortality (YPLL rate) – among 40-49 age group
32. Trends in premature mortality (YPLL rate) – among 50-59 age group
33. Trends in premature mortality (YPLL rate) – among 60-69 age group

Diabetes & Endocrine Disorders

34. Trends in premature mortality (YPLL rate) – overall
35. Trends in premature mortality (YPLL rate) – among females
36. Trends in premature mortality (YPLL rate) – among males
37. Trends in premature mortality (YPLL rate) – among 0-14 age group
38. Trends in premature mortality (YPLL rate) – among 15-29 age group
39. Trends in premature mortality (YPLL rate) – among 30-39 age group
40. Trends in premature mortality (YPLL rate) – among 40-49 age group
41. Trends in premature mortality (YPLL rate) – among 50-59 age group
42. Trends in premature mortality (YPLL rate) – among 60-69 age group

Circulatory diseases

43. Trends in premature mortality (YPLL rate) – overall
44. Trends in premature mortality (YPLL rate) – among females
45. Trends in premature mortality (YPLL rate) – among males
46. Trends in premature mortality (YPLL rate) – among 0-14 age group
47. Trends in premature mortality (YPLL rate) – among 15-29 age group
48. Trends in premature mortality (YPLL rate) – among 30-39 age group
49. Trends in premature mortality (YPLL rate) – among 40-49 age group
50. Trends in premature mortality (YPLL rate) – among 50-59 age group
51. Trends in premature mortality (YPLL rate) – among 60-69 age group

Chronic Respiratory Diseases

Analysis Plan

52. Trends in premature mortality (YPLL rate) – overall
53. Trends in premature mortality (YPLL rate) – among females
54. Trends in premature mortality (YPLL rate) – among males
55. Trends in premature mortality (YPLL rate) – among 0-14 age group
56. Trends in premature mortality (YPLL rate) – among 15-29 age group
57. Trends in premature mortality (YPLL rate) – among 30-39 age group
58. Trends in premature mortality (YPLL rate) – among 40-49 age group
59. Trends in premature mortality (YPLL rate) – among 50-59 age group
60. Trends in premature mortality (YPLL rate) – among 60-69 age group

Digestive diseases

61. Trends in premature mortality (YPLL rate) – overall
62. Trends in premature mortality (YPLL rate) – among females
63. Trends in premature mortality (YPLL rate) – among males
64. Trends in premature mortality (YPLL rate) – among 0-14 age group
65. Trends in premature mortality (YPLL rate) – among 15-29 age group
66. Trends in premature mortality (YPLL rate) – among 30-39 age group
67. Trends in premature mortality (YPLL rate) – among 40-49 age group
68. Trends in premature mortality (YPLL rate) – among 50-59 age group
69. Trends in premature mortality (YPLL rate) – among 60-69 age group

Genitourinary Diseases

70. Trends in premature mortality (YPLL rate) – overall
71. Trends in premature mortality (YPLL rate) – among females
72. Trends in premature mortality (YPLL rate) – among males
73. Trends in premature mortality (YPLL rate) – among 0-14 age group
74. Trends in premature mortality (YPLL rate) – among 15-29 age group
75. Trends in premature mortality (YPLL rate) – among 30-39 age group
76. Trends in premature mortality (YPLL rate) – among 40-49 age group
77. Trends in premature mortality (YPLL rate) – among 50-59 age group
78. Trends in premature mortality (YPLL rate) – among 60-69 age group

Neuropsychiatric Conditions

79. Trends in premature mortality (YPLL rate) – overall
80. Trends in premature mortality (YPLL rate) – among females
81. Trends in premature mortality (YPLL rate) – among males
82. Trends in premature mortality (YPLL rate) – among 0-14 age group
83. Trends in premature mortality (YPLL rate) – among 15-29 age group
84. Trends in premature mortality (YPLL rate) – among 30-39 age group
85. Trends in premature mortality (YPLL rate) – among 40-49 age group
86. Trends in premature mortality (YPLL rate) – among 50-59 age group
87. Trends in premature mortality (YPLL rate) – among 60-69 age group

External Causes

88. Trends in premature mortality (YPLL rate) – overall
89. Trends in premature mortality (YPLL rate) – among females
90. Trends in premature mortality (YPLL rate) – among males
91. Trends in premature mortality (YPLL rate) – among 0-14 age group
92. Trends in premature mortality (YPLL rate) – among 15-29 age group
93. Trends in premature mortality (YPLL rate) – among 30-39 age group
94. Trends in premature mortality (YPLL rate) – among 40-49 age group
95. Trends in premature mortality (YPLL rate) – among 50-59 age group
96. Trends in premature mortality (YPLL rate) – among 60-69 age group

Other Causes

Analysis Plan

97. Trends in premature mortality (YPLL rate) – overall
98. Trends in premature mortality (YPLL rate) – among females
99. Trends in premature mortality (YPLL rate) – among males
100. Trends in premature mortality (YPLL rate) – among 0-14 age group
101. Trends in premature mortality (YPLL rate) – among 15-29 age group
102. Trends in premature mortality (YPLL rate) – among 30-39 age group
103. Trends in premature mortality (YPLL rate) – among 40-49 age group
104. Trends in premature mortality (YPLL rate) – among 50-59 age group
105. Trends in premature mortality (YPLL rate) – among 60-69 age group

- Joinpoint models will use a log-linear specification with calculated annual standard errors supplied to account for heteroscedasticity. Autocorrelation will be set to zero. The weighted Bayesian Information Criterion will be used to select between models with zero and one joinpoint, with the maximum restricted to one. APCs and AAPCs will be reported with 95% confidence intervals.
- Other Analyses
    1. Chi squared test to assess differences in the proportion of premature deaths by sex, for all causes and by each cause, in 2010 and 2019 (α=0.05).
        - Apply Bonferroni correction by multiplying each p-value by the total number of comparisons conducted (cause categories x years of comparison)
    2. Mann Whitney U/Wilcoxon Rank Sum test to assess differences in the median years of potential life lost per premature death by sex, for all causes and by each cause, in 2010 and 2019 (α=0.05).
        - Apply Bonferroni correction by multiplying each p-value by the total number of comparisons conducted (cause categories x years of comparison)
    3. Estimate Poisson regression models for the association between sex and (1) the rate of premature mortality and (2) the rate of YPLL using data aggregated by five-year age group, sex, and calendar year. Models will include population size as the exposure, use robust standard errors, and adjust for age group and calendar year
    4. As a supplementary validation, RGD-recorded homicide deaths will be compared with JCF-reported homicide victim counts for 2013-2019, restricted to ages 0-64 years, to assess potential under-registration of violent deaths in the vital registration data.
        - Strict definition: ICD-10 X85-Y09, Y87.1
        - Expanded definition: ICD-10 X85-Y09, Y87.1, Y22-Y24, Y28-Y29
        - The percentage of JCF-reported homicide victims captured by the RGD will be calculated overall for each year, and by age group pooled over the entire study period.
        - Age-standardized YPLL rates will also be calculated for JCF-reported homicides, and temporal trends will be assessed using Joinpoint regression.

Appendix

Diagnosis Codes for Cause of Death Groupings	
Cause	**Diagnosis Codes**
Maternal & perinatal conditions	O00-O999 P00-P229 P24-P369 P370 P371 P372 P375 P378 P379 P38-P969
Infectious & parasitic diseases	A00-B999 G00-G049 G14* H65-H669 J00-J22 N70-N739 P23* P373 P374 U04* U071 U072 U099 U109
Cancer	C00-C979
Diabetes and endocrine disorders	E10-E149 D55-D639 D640 D641 D642 D643 D644 D648 D65-D899 E03-E079 E15-E169 E20-E349 E65-E889
Circulatory Diseases	I00-I999
Chronic Respiratory Diseases	J30-J989
Oral & Digestive Diseases	K00-K929
Genitourinary Diseases	N00-N649 N75-N989
Neuropsychiatric Conditions	F01-F999 G06-G139 G20-G989 U070 X41* X42* X44* X45*
External Causes	V01-X409 X43* X46-X999 Y00-Y899 U129
Other Causes	All other diagnosis codes not captured above, including benign neoplasms, musculoskeletal diseases, sense organ diseases, skin diseases, nutritional deficiencies, and congenital anomalies. Diagnosis codes sourced and categorizations adapted from: https://platform.who.int/mortality/about/list-of-causes-and-corresponding-icd-10-codes